



NEXUS-1

ROOT-CAUSE ANALYSIS ENGINE

Implementation roadmap for an effective diagnostic system

Phased stages with work-hour estimates · no code

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Build signature NX1-094C79E0A366D34F

June 2026 · Planning document

ADVISORY DIAGNOSTIC · ESTIMATE, NOT A QUOTE

The engine informs operators and investigators. It never feeds control or protection actions.

Hours are planning estimates and assume a hybrid expert/data model with significant domain-expert input.

Scope & approach

This roadmap turns the NEXUS-1 **Root Cause Graph** demonstration into a credible, validated diagnostic engine. The demo already proves the core idea: an engineered fault-propagation backbone lit by live alarms, the ability to test any node as the root cause, an alarm-flood scenario that collapses many cascading alarms to a single origin, and a ranking that contrasts causal-origin reasoning with naive alarm-counting. The work below is what stands between that demonstration and a system an operator or investigator could rely on.

Approach taken in this plan (the agreed defaults):

- **Hybrid knowledge model** — an expert-built causal backbone (fault trees, FMEA, cause-and-effect matrices) with weights and delays tuned from historian data. Not a hand-built graph alone, and not an opaque learned model.
- **Both modes** — a real-time engine that guides operators inside an alarm flood, and an investigative mode engineers use to reconstruct events after the fact.
- **Advisory only** — it informs people; it never feeds control or protection. This shapes the validation and human-factors work.
- **Standalone roadmap** — readable on its own, but it references the wider platform (Kafka stream, alarm / incident / audit / dependency services) where it would plug in. It is the deep dive behind the platform roadmap's 'Incident & Root-cause service' line.
- **Format** — phased stages, each with a work-breakdown and Low / High work-hours; the Planning figure is the midpoint plus ~15% contingency. No code.

Two honest warnings up front:

- Diagnostic quality is capped by alarm quality (R1) and proven only by validation (R8). These two stages, not the graph itself, decide whether the engine is effective — budget and respect them accordingly.
- A large share of this effort — the event corpus, alarm rationalisation, the causal model, the independent review — is **domain knowledge engineering by reliability / process experts**, not software development. The hours include that, but it needs the right people, not just engineers.

From the demo to an engine

What the demonstration shows today, and what each stage adds to make it effective:

Aspect	In the demo today	What the engine adds
Causal model	Hand-built fault backbone	Governed, versioned, data-tuned causal graph (R2, R5)
Alarm input	Live alarms light nodes	Rationalised, de-noised, prioritised stream — EEMUA / ISA (R1)
Finding the cause	Operator tests a node as root	Automatic ranked hypotheses with confidence (R3, R4)
Reasoning basis	Topology + fixed delays	Temporal reasoning over alarm order + learned delays (R3.3, R5.1)
Alarm flood	Collapses ~14 alarms to one origin	Live, incremental collapse + consequential suppression (R6.4)
Ranking quality	Causal vs alarm-count contrast	Measured accuracy vs baselines on real events (R8)
What-if	A what-if toggle	Formal counterfactuals + post-event replay (R4.5, R7)

Effort summary

Stage	Name	Low	High	Planning
R0	Foundations & Data Readiness	185	315	290
R1	Alarm Rationalisation & Signal Conditioning	215	370	340
R2	Causal Knowledge Model (expert backbone)	300	520	470
R3	Inference Core — Topological + Temporal	260	450	410
R4	Uncertainty, Confidence & Ranking	250	445	400
R5	Data-Driven Tuning & Discovery (the hybrid layer)	235	415	370
R6	Real-time Engine & Operator Experience	280	480	440
R7	Investigative / Post-event Mode	180	320	290
R8	Validation, Verification & Tuning	260	455	410
R9	Integration, Operations & Lifecycle	220	380	340
PM	Programme management & technical leadership	320	470	450
ENGINE TOTAL		2,705	4,620	4,210

Engineering work-hours. Planning = midpoint + ~15% contingency. A substantial share is domain-expert effort (see R0, R1, R2, R8).

Indicative calendar (planning hours)

Team	Composition	Approx. duration
Focused (2-3)	1-2 engineers + part-time data scientist + part-time reliability SME	~13 months to a validated v1
Solo + SME	1 engineer with periodic SME support	~2+ years — not advised for an effective system

Stages R3-R5 (inference, confidence, tuning) overlap once the model and conditioned stream exist; R6 and R7 share most of the engine and can be built in sequence. Duration assumes the platform's alarm and event infrastructure already exists.

Milestones

Milestone	Stages	What you can trust at this point
M1 — Inputs ready	R0 + R1	A clean, prioritised event stream and a labelled corpus of past events to test against.
M2 — Reasoning works	R2 + R3	A governed causal model that produces ranked, timing-aware candidate causes with explanations.
M3 — Trustworthy answers	R4 + R5	Calibrated confidence, robustness to messy alarms, and parameters grounded in plant data.
M4 — In the hands of users	R6 + R7	A live operator advisory inside the flood, and an investigative tool for after the event.
M5 — Validated v1	R8 + R9	Measured accuracy that beats naive baselines, integrated and under change-control governance.

Recommended first step: build M1 and a thin slice of M2-M3 against a handful of corpus events before committing to the full model. Proving the reasoning on three real events de-risks the entire estimate.

Detailed stages (work breakdown)

Foundations & Data Readiness

R0

Decide what 'good diagnosis' means and assemble the inputs the engine will reason over. The most valuable artefact of this stage is a curated library of past events with agreed root causes — the yardstick everything later is measured against.

ID	Step	Low	High	Notes
R0.1	Diagnostic objectives & success metrics	25	45	Correct-root rate, time-to-diagnosis, false-attribution
R0.2	Data-source inventory & access	30	50	Alarms, historian, asset registry, dependency graph
R0.3	Event / alarm data model + time-sync requirements	25	45	Ordering integrity is essential for temporal logic
R0.4	Curate historical event corpus (gold standard)	60	100	SME-led; the validation yardstick
R0.5	Stakeholder alignment & operating-context capture	25	40	Operators, reliability, safety
R0.6	Baseline today's diagnostic performance	20	35	What good looks like vs current practice
Stage subtotal		185	315	Planning ~290 h

- › Roughly a third of this whole roadmap is knowledge engineering by domain experts, not software. R0.4 sets that tone.
- › Output: clear targets and a labelled event corpus to validate against.

Alarm Rationalisation & Signal Conditioning

R1

The unglamorous layer that decides everything: diagnostic quality cannot exceed alarm quality. Clean, prioritised, de-noised events in; nonsense in means nonsense out. Aligned to EEMUA 191 / ISA-18.2 alarm-management practice.

ID	Step	Low	High	Notes
R1.1	Alarm rationalisation (EEMUA 191 / ISA-18.2)	60	100	Priorities, setpoints, deadbands — SME-led
R1.2	Chattering / fleeting alarm suppression + debounce	35	60	
R1.3	Standing / stale / shelved alarm handling	30	55	
R1.4	Alarm grouping, first-out / first-up logic	35	60	
R1.5	Alarm-performance metrics & monitoring	30	50	Flood rate, standing count, EEMUA KPIs
R1.6	Conditioned event-stream contract (engine input)	25	45	The clean feed the engine consumes
Stage subtotal		215	370	Planning ~340 h

- › This stage is where many real diagnostic projects quietly fail; treat it as first-class, not plumbing.
- › Output: a rationalised, de-noised, prioritised event stream.

Causal Knowledge Model (expert backbone)

R2

Formalise the plant's cause-and-effect knowledge into a governed, versioned causal graph — the principled successor to the demo's hand-built fault backbone. Extends, rather than discards, what the demo proves.

ID	Step	Low	High	Notes
R2.1	Causal-graph schema (nodes, failure modes, symptoms, edges)	40	70	Edges carry delay ranges + conditions
R2.2	Formalise fault trees / FMEA / cause-effect matrices	90	150	Largest SME effort in the programme
R2.3	Signal-to-node binding layer	50	80	Map alarms / tags to model nodes
R2.4	Propagation-delay & condition library	40	70	Timing windows per causal edge
R2.5	Model versioning, governance & coverage analysis	40	70	Which failure modes are covered
R2.6	Model authoring / maintenance tooling	40	80	Engineers edit the model without code
Stage subtotal		300	520	Planning ~470 h

- › The model must be maintained as the plant changes; governance (R2.5) is not optional.
- › Output: a maintainable, governed causal graph with measured coverage.

Inference Core — Topological + Temporal

R3

The reasoning engine. Beyond the demo's 'test a node as root', it generates and ranks candidate causes, and — critically — uses alarm timing, not just graph topology, to tell true causes from coincidences.

ID	Step	Low	High	Notes
R3.1	Forward / backward propagation engine	50	80	Generalises the demo's node test
R3.2	Hypothesis generation (roots that explain the alarm set)	50	90	
R3.3	Temporal reasoning (arrival order + delay windows)	70	120	The key differentiator vs topology-only
R3.4	Hypothesis scoring (coverage, timing fit, parsimony)	50	90	
R3.5	Explanation generation (why this root, which alarms)	40	70	Explainability is a requirement
Stage subtotal		260	450	Planning ~410 h

- › A cause must precede its effects within the modelled delay windows — timing turns a plausible graph path into a defensible diagnosis.
- › Output: ranked candidate root causes, each with an explanation.

Uncertainty, Confidence & Ranking

R4

Real plants give incomplete, late and spurious alarms, and faults can coincide. The engine must offer ranked hypotheses with calibrated confidence and the evidence for and against each — not a single brittle answer.

ID	Step	Low	High	Notes
R4.1	Probabilistic inference layer (Bayesian / evidential)	80	140	Scores become calibrated confidence
R4.2	Noise robustness (missing / late / spurious alarms)	50	90	
R4.3	Multiple / simultaneous & common-cause faults	50	90	
R4.4	Confidence calibration	40	70	Stated confidence matches observed accuracy
R4.5	Counterfactual / what-if formalisation	30	55	Generalises the demo's what-if toggle
Stage subtotal		250	445	Planning ~400 h

- › Output: ranked, calibrated, explainable hypotheses robust to imperfect inputs.

Data-Driven Tuning & Discovery (the hybrid layer)

R5

Ground the expert model in data: learn edge weights and delays from the historian, surface candidate relationships for expert review, and detect when the plant has drifted from the model. Humans approve every model change.

ID	Step	Low	High	Notes
R5.1	Edge-weight & delay learning from historian	70	120	
R5.2	Data-driven candidate-edge suggestions (for review)	60	110	Suggest, never auto-commit
R5.3	Model drift detection	40	70	Flags when reality diverges from model
R5.4	Operator-feedback active-learning loop	40	70	Confirmed causes refine the model
R5.5	Human-in-the-loop change approval & governance	25	45	
Stage subtotal		235	415	Planning ~370 h

- › Hybrid = expert backbone, data-tuned parameters, human-governed changes. Never an opaque learned black box.
- › Output: a data-grounded model whose evolution stays under expert control.

Real-time Engine & Operator Experience

R6

Run continuously on the conditioned stream and surface a probable cause within the alarm flood, as it happens. Incremental — it updates per alarm rather than recomputing — and presented as calm, clearly advisory guidance.

ID	Step	Low	High	Notes
R6.1	Online inference pipeline (consume conditioned stream)	70	120	Kafka input from the platform
R6.2	Incremental inference + latency targets	60	100	Update per alarm, not from scratch
R6.3	Operator UI: live ranked root cause + lit graph	70	120	Extends the demo's graph & ranking
R6.4	Alarm-flood collapse + consequential-alarm suppression	50	90	The demo's 14-to-1 collapse, made live
R6.5	Human-factors review + advisory framing	30	50	Clear it informs, never acts
Stage subtotal		280	480	Planning ~440 h

- › Output: a live, incremental, operator-facing diagnostic.

Investigative / Post-event Mode

R7

The engineer-facing counterpart: load a historical window, replay it, and produce a defensible root-cause report with timeline and evidence — plus 'what would have prevented this' re-analysis. Feeds incident and audit records.

ID	Step	Low	High	Notes
R7.1	Offline event replay & reconstruction	60	100	
R7.2	Causal-chain reconstruction + timeline	50	90	Generalises the demo's trip reconstruction
R7.3	Root-cause report generation & export	40	70	To incident / audit records
R7.4	What-if re-analysis (prevention)	30	60	
Stage subtotal		180	320	Planning ~290 h

- › Output: a post-event investigation tool with exportable, evidenced reports.

Validation, Verification & Tuning

R8

Where 'effective' is proven or disproven. Back-test the engine against the curated corpus, measure accuracy honestly, and demonstrate it beats the naive baselines the demo already contrasts against (count / recency / frequency).

ID	Step	Low	High	Notes
R8.1	Back-test harness against the historical corpus	60	100	Uses the R0.4 gold standard
R8.2	Accuracy metrics (root rate, rank-of-truth, time, false-attribution)	50	90	
R8.3	Benchmark vs naive baselines (count / recency / frequency)	30	55	The demo's count-vs-causal contrast, measured
R8.4	Sensitivity & coverage-gap analysis	50	90	
R8.5	Independent model review + acceptance sign-off	70	120	SME / independent — credibility gate
Stage subtotal		260	455	Planning ~410 h

- › Publish the numbers, including the misses. A diagnostic engine without measured accuracy is a demo, not a tool.
- › Output: a measured, evidence-backed engine with documented accuracy and limits.

Integration, Operations & Lifecycle

R9

Make it operable and keep it honest over time: integrate with the platform services, log every diagnosis with its evidence and model version, and put a change-control process around the causal model so it tracks the real plant.

ID	Step	Low	High	Notes
R9.1	Platform integration (alarms / incident / audit / dependency)	60	100	Shares topology with the dependency service
R9.2	Engine observability + per-diagnosis provenance	50	90	Which model version produced which call
R9.3	Model governance lifecycle + periodic re-validation	40	70	
R9.4	Management-of-change process for the causal model	30	50	
R9.5	Operator / engineer training & documentation	40	70	
Stage subtotal		220	380	Planning ~340 h

- › Output: an operable, governed, maintainable diagnostic capability.

Relationship to the platform & caveats

- This engine is the depth behind the platform roadmap's 'Incident & Root-cause service'. It assumes that roadmap's alarm service, event/Kafka pipeline, historian and dependency-graph service exist — it consumes the conditioned alarm stream and shares plant topology with the dependency service rather than rebuilding either.
- Estimates are planning figures, not a fixed-price quote, and assume senior engineers plus genuine reliability / process SME availability. The dominant risks are the quality of the alarm rationalisation (R1) and the realism of the causal model (R2); both are mitigated by the validation stage (R8).
- The engine is advisory throughout. It supports operators and investigators and produces evidenced, explainable diagnoses; it is never wired to control or protection actuation, which remain hardware-isolated and human-supervised.
- Re-validate the estimate after the recommended first slice (M1 + a thin M2-M3 on three real events). Diagnostic accuracy on real data, not graph aesthetics, is the only meaningful measure of success.

This roadmap and the NEXUS-1 console are the original work of Grigorios Agathangelidis. Build signature NX1-094C79E0A366D34F.